

Slides Notes

2017.00

This section covers the history of computing and the theoretical foundations of interface design.

Hardware evolution: From abacus (16th century) and mechanical machines to vacuum tubes and transistors.

Key pioneers: Jacquard (first programmable loom), Charles Babbage, Ada Lovelace, and Hollerith.

The Human Interface (Rashid's laws):

A computer shall not harm your work.

A computer shall not waste your time.

An interface is humane if it is responsive to human needs and facilitates the user should set the pace of interaction.

2017.01

Locus of Attention: Humans generally have a single continuous focus of attention.

Shifting Locus: It takes a measurable amount of time to shift focus. Interfaces should minimize unnecessary distractions.

Absorption: Design assuming the user will be absorbed in their work and might not notice peripheral alerts.

- Getting started with UI Design

Interface Metaphors: Using familiar environment (e.g. Desktop, Control Panel, Store) to help users understand the system.

Idioms: Standardized interaction patterns that are learned once applied in many contexts.

User-Centric Design: Considering user motivation, ability, needs, and behavioral patterns from the start.

2.07.02

Organizational Models: Ways to structure data so it makes sense:
Linear, Tabular, Hierarchical (trees), Network, Geographic

Goal: To present data visually to impart knowledge using graphics, tables and maps.

2.07.03

Getting started with user research.

- Understand user classes, their goals, language, and attitudes.
- Direct observation, interviews, case studies, surveys and "Personas"
- Solve the right problem and ask "why" repeatedly.

Visual hierarchy: Guiding the users' eye through grouping and alignment

Gestalt principles: Proximity, Similarity, Continuity, Closure

- Visual Framework, center stage
- Titled Sections
- Prioritizing
- Isolated layout.

2.07.04

Practical examples of requirements and scenarios

- author scenario

Dr. Howard:

- registration
- activation
- login
- uploading a PDF
- submitting for review

reviewer Scenarios

- list of assigned papers.
- selecting one
- reading
- adding comments
- and voting

Editing Scenarios: Editing comments and changing a vote.

Context: "Flight" a tool for modeling and optimizing traffic at inter sections (lanes, lights).

Functional requirements

users

- Cartographers (drawing maps)
- Traffic Engineers (creating simulations)
- Screen for simulation execution.
- Simulation Data viewing.
- Map selection
- Analysis.

2.02.06

Mechanisms:

- scroll/pan
- zoom
- drill down
- Magnify.

Way finding: Helping the user understand where they are and how to get to where they want to go

Data manipulation

- Sorting (Alphabetically, numerically).
- Filtering and querying.

2.02.07

Goal: Make actions discoverable and easy to execute

Button groups: Grouping related actions together visually

Action Panel: A dedicated area for commands (like a sidebar) that contextualizes changes

Smart Menus: Menus that adapt or show the most used items (though this can confuse users if not done well).

Undo Redo: Critical for allowing users to explore without fear of

Progress indicators: Essential for long-running actions to reassure the user the system is working.

2.07.19

Showing Complex Data.

Data Density

Core Goal:

- Present data visually to impart knowledge (High Data - Enticement)
- Avoid overwhelming the user while providing depth

Organizational Models.

- Linear
- Tabular
- Hierarchical
- Network
- Geographic

Visual Patterns and Techniques

Tree table

- Combine hierarchy (expanded tree) with detailed columns

Dynamic Views.

- Allows users to reconfigure data (Sort, Filter, Query).

Overview + Detail.

- Data Hips / Tooltips: Show small details on hover to keep the view clean
- Master Detail: Selecting an item in a list shows its full details in a separate pane.

Coordinated Views

Selecting an item in one visualization automatically highlights the corresponding data in another view

02.07.11

Core Mechanism Form.

- the primary interface for data entry
- Should map to the user's mental model.

UI Clarity Principles

Good Defaults

- Pre-fill fields with the most likely values to save typing.

Forgiving Format

- allow flexible input

Input hints

- placeholder text or examples inside

Feedback and validation

inline validation

- Report errors immediately after the user leaves a field, rather than waiting for the "submit" button.

Actionable Error Messages

- Explain what is wrong and how to fix it (avoid generic "invalid input" errors)